

No Association of Coffee Consumption with Gastric Ulcer, Duodenal Ulcer, Reflux Esophagitis, and Non-Erosive Reflux Disease: A Cross-Sectional Study of 8,013 Healthy Subjects in Japan

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Probably due to caffeine-induced gastric acid secretion, negative effects of coffee upon various upper-gastrointestinal diseases have been precariously accepted, despite the inadequate epidemiological evidence. Our aim is to evaluate the effect of coffee consumption on four major acid-related diseases: gastric ulcer (GU), duodenal ulcer (DU), reflux esophagitis (RE), and non-erosive reflux disease (NERD) based on the large-scale multivariate analysis. Of the 9,517 healthy adults, GU, DU, and RE were diagnosed by endoscopy, and NERD was diagnosed by the symptoms of heartburn and regurgitation without esophageal erosion. Associations between coffee consumption and the four disorders were evaluated, together with age, gender, body mass index (BMI), *Helicobacter pylori* (HP) infection status, pepsinogen I/II ratio, smoking, and alcohol. We further performed meta-analysis using the random effects model to redefine the relationship between coffee intake and peptic ulcer disease. The eligible 8,013 study subjects comprised of 5,451 coffee drinkers and 2,562 non-coffee drinkers. By univariate analysis, age, BMI, pepsinogen I/II ratio, smoking, and alcohol showed significant associations with coffee consumption. By multiple logistic regression analysis, positively correlated factors with significance were HP infection, current smoking, BMI, and pepsinogen I/II ratio for GU; HP infection, pepsinogen I/II ratio, and current smoking for DU; HP non-infection, male, BMI, pepsinogen I/II ratio, smoking, age, and alcohol for RE; younger age, smoking, and female for NERD. The meta-analyses could detect any association of coffee consumption with neither GU nor DU. In conclusion, there are no significant relationship between coffee consumption and the four major acid-related upper gastrointestinal disorders.